PHASE 1 & PHASE 2 CONSTRUCTION COST

Patel Mansion - 30×60 G+1, Bhabua, Bihar

Document Date: March 13, 2026
Project Owner: Harsh
Total Budget: ₹85,00,000
Phase 1 + 2 Cost: ₹37,89,699 (44.6% of budget)

EXECUTIVE SUMMARY

This document provides a complete, material-level breakdown of **Phase 1 (Foundation & Structure)** and **Phase 2 (Walls, Plaster & Doors)** for Patel Mansion construction project in Bhabua, Bihar.

Quick Overview

Phase	Scope	Cost	% of Total Budget
Phase 1	Foundation, columns, beams, slabs	₹29,84,200	35.1%
Phase 2	Walls, plaster, 11 doors	₹8,05,499	9.5%
Combined	**Complete structural shell**	**₹37,89,699**	**44.6%**

Key Specifications

- 30 footings (4×4 ft, 5 ft deep, M25 concrete)
- 30 columns (9×12 inches, Fe500D steel)
- Beams (9×18 inches, M20 concrete)
- Slabs (5 inches thick, M20 concrete)

- UltraTech PPC cement only
- Tata Tiscon Fe500D steel only
- AAC blocks for outer walls, fly ash bricks for inner walls
- 11 premium doors with Atomberg smart lock on main entrance

PHASE 1: FOUNDATION & STRUCTURE

Total Cost: ₹29,84,200

Phase 1 creates the complete RCC skeleton from ground excavation to roof-level structural completion.

Material-wise Breakdown

Material	Quantity	Rate	Cost
Steel (Tata Tiscon Fe500D)	16.1 MT	₹62,000/MT	₹9,98,200
Cement (UltraTech PPC)	1,100 bags	₹400/bag	₹4,40,000
Sand (Son/Kaimur Zone II)	2,200 cft	₹55/cft	₹1,21,000
Crushed Aggregate (20mm)	3,400 cft	₹60/cft	₹2,04,000
Shuttering & Centering	4,500 sq ft	₹104/sq ft	₹4,68,000
RCC Labour	Lump sum	---	₹4,28,000
Excavation & Backfilling	Lump sum	---	₹1,40,000
Consumables & Hardware	Lump sum	---	₹1,85,000
TOTAL			**₹29,84,200**

Table 1: Phase 1 Material-wise Cost Distribution

Stage-wise Breakdown

Stage	Description	Cost
Pre-start	Site cleaning, layout marking, temp water/power	₹45,000
Excavation	Pit excavation for 30 footings, soil disposal, leveling	₹1,55,000
Foundation base	PCC bed material, labour, anti-termite treatment	₹1,18,000
Footings	RCC footings (30 nos), steel, shuttering, M25 concrete	₹4,65,000
Pedestal/neck	Column starters, pedestal steel, concrete, labour	₹80,000
Plinth beam	Beam steel, shuttering, M20 concrete, backfilling	₹2,57,000
Columns	Column steel, shuttering, M20 concrete (30 nos)	₹5,06,000
Beams	Beam steel, shuttering/centering, M20 concrete	₹7,20,000
Slabs	Slab steel, centering, M20 concrete, labour, curing	₹8,13,000
General RCC	Mixer/vibrator hire, testing, scaffolding, wastage	₹2,39,000
PHASE 1 TOTAL		**₹29,84,200**

Table 2: Phase 1 Stage-wise Cost Distribution

Phase 1 Cost Drivers

The three largest cost components in Phase 1:

1. **Steel (Tata Tiscon Fe500D):** ₹9,98,200 (33.4% of Phase 1) — 16.1 metric tons total
2. **Concrete Materials (Cement/Sand/Aggregate):** ₹7,65,000 (25.6% of Phase 1)
3. **Shuttering & Centering:** ₹4,68,000 (15.7% of Phase 1)

Quality Control Checkpoints

- Verify steel diameter, spacing, and cover blocks before each RCC pour
- Insist on proper curing (14 days minimum for columns/beams, 28 days for slabs)
- Do not allow contractor to substitute Tata Tiscon Fe500D or UltraTech PPC
- Monitor shuttering alignment and level before concrete casting
- Track actual steel consumption against 16.1 MT budget
- Track actual cement consumption against 1,100 bags budget

PHASE 2: WALLS, PLASTER & DOORS

Total Cost: ₹8,05,499

Phase 2 transforms the bare RCC skeleton into a fully enclosed shell with all doors installed.

Complete Item Breakdown

Item	Description	Specification	Cost
AAC outer walls	External perimeter walls	2,800 blocks @ ₹50	₹1,40,000
AAC labour	Masonry labour for AAC	Lump sum	₹45,000
Fly ash inner walls	Internal partitions	18,000 bricks @ ₹9	₹1,62,000
Fly ash labour	Masonry labour for bricks	Lump sum	₹78,000
Internal plaster	1:4 mix, 2 coats	~280 bags cement + sand	₹1,25,000
External plaster	1:6 mix + waterproofing	~180 bags cement + sand	₹95,000
Main door	Solid Teak 4-panel 7.5×3.5 ft	Atomberg smart lock	₹37,899
Bedroom doors (3)	Walnut groove laminate 7.5×3 ft	BR1, BR2, Guest	₹40,800
Kitchen door	Ivory gloss flush 7×3 ft	Hettich matte black	₹12,600
Bathroom doors (3)	WPC waterproof 7×2.5 ft	Godrej NEH	₹28,200
Puja door	Frosted glass panel 7×2.5 ft	Hettich satin gold	₹10,200
Store door	Basic hollow core 7×2.5 ft	Basic hardware	₹2,800
Door frames & fitting	Sal wood frames, hinges	11 doors total	₹28,000
PHASE 2 TOTAL			**₹8,05,499**

Table 3: Phase 2 Complete Item-wise Breakdown

Phase 2 Summary by Category

Category	Amount
AAC Blocks (Outer walls)	₹1,85,000
Fly Ash Bricks (Inner walls)	₹2,40,000
Plaster (Internal + External)	₹2,20,000
Doors (11 nos with hardware)	₹1,60,499
PHASE 2 TOTAL	**₹8,05,499**

Table 4: Phase 2 Category-wise Summary

Door Specifications (11 Doors Total)

Main Entrance Door:

- Material: Solid Teak 4-panel design
- Size: 7.5×3.5 ft (premium height)
- Frame: Sal wood
- Hardware: Ozone D-pull, Atomberg Azhero smart lock (fingerprint/PIN/app/NFC), Hettich hinges
- Cost: ₹37,899

Bedroom Doors (BR1, BR2, Guest - 3 nos):

- Material: Blockboard with Walnut groove laminate
- Size: 7.5×3 ft
- Frame: Sal wood
- Hardware: Hettich lever matte black, Dorset lock, Hettich hinges
- Cost: ₹40,800 total (₹13,600 each)

Kitchen Door:

- Material: Blockboard Ivory gloss flush
- Size: 7×3 ft
- Hardware: Hettich lever matte black, Dorset lock
- Cost: ₹12,600

Bathroom Doors (3 nos):

- Material: WPC (Wood Plastic Composite) - 100% waterproof
- Size: 7×2.5 ft
- Frame: WPC frame

- Hardware: Godrej NEH lever matte black, turn-knob lock
- Cost: ₹28,200 total (₹9,400 each)

Puja Room Door:

- Material: Flush with frosted glass panel
- Size: 7×2.5 ft
- Hardware: Hettich satin gold handle
- Cost: ₹10,200

Store Room Door:

- Material: Basic hollow core flush
- Size: 7×2.5 ft
- Hardware: Basic handle and lock
- Cost: ₹2,800

Phase 2 Material Requirements

Cement for Plaster:

- Internal plaster (1:4 mix): ~280 bags UltraTech PPC
- External plaster (1:6 mix): ~180 bags UltraTech PPC
- Total plaster cement: ~460 bags

Sand for Plaster:

- Total sand requirement: ~1,100 cubic feet (Son/Kaimur Zone II)

Wall Materials:

- AAC blocks (outer walls): ~2,800 blocks @ ₹50/block
- Fly ash bricks (inner walls): ~18,000 bricks @ ₹9/brick

COMBINED PHASE 1 + PHASE 2 ANALYSIS

Cost Summary

Phase	Cost	% of ₹85L Budget
Phase 1 - Foundation & Structure	₹29,84,200	35.1%
Phase 2 - Walls, Plaster & Doors	₹8,05,499	9.5%
Combined Total	**₹37,89,699**	**44.6%**
Remaining budget for Phases 3-8	₹47,10,301	55.4%

Table 5: Combined Phase 1 + 2 Budget Analysis

Budget Health Check

At ₹37.89 lakh for Phase 1 + 2, you have consumed **44.6% of your ₹85 lakh total budget** with the structural shell complete.

Remaining phases:

- Phase 3: Windows & Glazing
- Phase 4: Tiles & Flooring
- Phase 5: Electrical & Plumbing
- Phase 6: Paint & False Ceiling
- Phase 7: Kitchen & Wardrobes
- Phase 8: External Works & Overheads

Remaining budget: ₹47,10,301 (55.4%)

This is a healthy position. The structural foundation represents the largest single cost block, and you are tracking within budget targets.

Payment Milestones

Phase 1 Payment Schedule:

Milestone	Work Complete	Payment
Release 1	Excavation + PCC bed complete	₹2,50,000
Release 2	Footings + plinth beam complete	₹7,00,000
Release 3	Columns + beams complete	₹9,00,000
Release 4	All slabs complete	₹11,34,200
Phase 1 Total		**₹29,84,200**

Table 6: Phase 1 Milestone-based Payment Schedule

Phase 2 Payment Schedule:

Milestone	Work Complete	Payment
Release 1	AAC + fly ash walls complete	₹4,25,000
Release 2	Internal + external plaster complete	₹2,20,000
Release 3	All 11 doors fitted with hardware	₹1,60,499
Phase 2 Total		**₹8,05,499**

Table 7: Phase 2 Milestone-based Payment Schedule

Critical Control Points

For Phase 1:

1. Never allow steel or cement brand substitution
2. Track actual consumption against budgeted quantities (16.1 MT steel, 1,100 bags cement)
3. Insist on proper curing duration (14-28 days)
4. Verify concrete grade before each pour (M25 for footings, M20 for rest)
5. Check shuttering alignment and level

For Phase 2:

1. Verify AAC block quality (lightweight, uniform size)
2. Check plaster mix ratio (1:4 internal, 1:6 external)
3. Confirm Atomberg smart lock model (Azhero series)
4. Verify WPC doors for bathrooms (100% waterproof)
5. Check all door hardware is matte black (except Puja - satin gold)

PROCUREMENT SCHEDULE

Phase 1 Material Procurement

Week 1-2 (Pre-start):

- Order steel: 16.1 MT Tata Tiscon Fe500D (bulk order from Patna dealer)
- Order cement: 1,100 bags UltraTech PPC
- Arrange sand: 2,200 cft Son/Kaimur Zone II

- Arrange aggregate: 3,400 cft 20mm crushed

Week 3-8 (Foundation work):

- Shuttering material rental/purchase
- Binding wire, cover blocks, consumables
- Concrete mixer and vibrator hire

Phase 2 Material Procurement

Week 9-11 (Walls):

- AAC blocks: 2,800 nos
- Fly ash bricks: 18,000 nos
- Cement for plaster: 460 bags UltraTech PPC
- Sand for plaster: 1,100 cft

Week 12-14 (Doors):

- Main door: Order Teak 4-panel + Atomberg Azhero lock
- Bedroom doors: Order Walnut groove laminate (3 nos)
- Kitchen door: Order Ivory gloss flush
- Bathroom doors: Order WPC waterproof (3 nos)
- Puja & Store doors: Order as per spec
- All hardware: Hettich, Godrej, Ozone

VENDOR RECOMMENDATIONS

Phase 1 Vendors

Material	Recommended Vendor
Steel (Tata Tiscon)	Bulk dealer in Patna (₹54-58/kg for 5+ MT orders)
Cement (UltraTech PPC)	Authorized dealer Bhabua/Patna
Sand	Son or Kaimur river suppliers
Aggregate	Mohania/Sasaram quarries
Shuttering	Local plywood dealers (Greenply/Century)

Table 8: Phase 1 Vendor Guide

Phase 2 Vendors

Item	Recommended Vendor
AAC blocks	Local AAC block manufacturers Bhabua/Patna
Fly ash bricks	Local brick kilns Bhabua
Teak door	Ozone dealers Patna (ozone.in)
Door hardware	Hettich authorized dealer Patna
Atomberg smart lock	Amazon India or atomberg.com
WPC doors	Godrej dealer (NEH series)

Table 9: Phase 2 Vendor Guide

TIMELINE ESTIMATE

Phase 1 Duration: 14 weeks (Weeks 1-14)

- Weeks 1-2: Site preparation, excavation, PCC
- Weeks 3-5: Footings and plinth beam
- Weeks 6-10: Columns and beams (GF + FF)
- Weeks 11-14: Slabs and curing

Phase 2 Duration: 6 weeks (Weeks 11-16, parallel with Phase 1)

- Weeks 11-13: Wall construction (AAC + fly ash)
- Weeks 14-15: Internal and external plaster
- Weeks 15-16: Door installation and hardware fitting

Combined Timeline: 16 weeks from site start to Phase 2 completion

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DOCUMENT CONTROL

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Date: March 13, 2026
Prepared By: Construction Planning Team

Usage:

- Hand this document to contractor for quotation reference
- Use for material procurement planning
- Track actual costs against budgeted amounts
- Monitor milestone completion for payment releases

Next Steps:

1. Confirm Phase 1 and Phase 2 specifications
2. Obtain contractor quotations against this detailed breakdown
3. Finalize vendor selection for steel and cement
4. Proceed with Phase 3 planning (Windows & Glazing)

All prices current as of March 2026. Obtain fresh local quotes before ordering.